

NAME :- B.Srikar

Ht.No:- 2303A51885

Question 1: Frog Jump

```
1 n = int(input())
2 hgts = list(map(int, input().split()))
3 dp = [0] * n
4 dp[0] = 0
5 dp[1] = abs(hgts[1] - hgts[0])
6
7 for i in range(2, n):
8     temp1 = dp[i - 1] + abs(hgts[i] - hgts[i - 1])
9     temp2 = dp[i - 2] + abs(hgts[i] - hgts[i - 2])
10    dp[i] = min(temp1, temp2)
11
12 print(dp[- 1])
```

Question 2: Frog Jump with K Distances

```
1 n, k = map(int, input().split())
2 hgts = list(map(int, input().split()))
3 dp = [float('inf')] * n
4 dp[0] = 0
5 for i in range(1, n):
6     for j in range(1, k + 1):
7         if i - j >= 0:
8             cost = dp[i - j] + abs(hgts[i] - hgts[i - j])
9             dp[i] = min(dp[i], cost)
10 |
11 print(dp[1])
```